

Souvik Mandol

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EDUCATION

Bangladesh University of Engineering and Technology

2022 - 2026

Bachelor of Science in Computer Science and Engineering

CGPA: 3.31/4.00

PROFESSIONAL EXPERIENCE

Junior Engineer (CSE), BRTC, Bangladesh University of Engineering and Technology

Aug 2026 - Present

- Government-funded national project (BRTC, BUET) building a locally developed adaptive traffic signal system for Dhaka city with Dhaka South City Corporation; live at 4 intersections, scaling to 30.
- Extend and maintain the Go and React/TypeScript platform (MQTT, PostgreSQL/TimescaleDB, WebSocket) that lets operators monitor and control signals from a tablet, with Telegram notifications on every change.
- Built the binary MQTT protocol layer and OTA firmware updates for STM32 Nucleo signal controllers, and the adaptive-mode pipeline that turns Jetson Orin Nano edge-vision traffic data into live, safety-clamped signal plans.
- Set up and run the on-premise server and private fiber network (IP addressing, Docker deployment) linking STM32 controllers, Jetson Orin Nano units, IP cameras and countdown timers, secured with per-device mTLS certificates and topic-level ACLs.

Software Engineer, MerilSoft

Dec 2025 - Apr 2026

- Developed an AI-driven magazine generation platform – [Magazine Works](#)
- Implemented file-handling systems on AWS and integrated AI-driven features using Google Vertex AI Studio.
- Worked on a restaurant system integrating existing online delivery platforms such as Uber Eats and DoorDash, along with a POS solution. - [Restaurant Solution](#)

PROJECTS

• Insomniyuck: Personal Blog

[Code](#)

Tools: SvelteKit, TypeScript, Cloudflare R2, Vercel

[Website](#)

- Insomniyuck is a personal memory archive built with SvelteKit and TypeScript, using mdsxv for Markdown-based content authoring. Features eight categorized sections spanning food, travel, running, books, movies, and personal reflections, with media assets served via Cloudflare R2 and an anonymous messaging integration.

• Art Gallery Management: A Database Project

[Code](#)

Tools: HTML, CSS, React, Javascript, NodeJS, ExpressJS, Oracle

- Art gallery management system built on Oracle and SQL with a React, Node.js and Express.js front end, handling artwork inventory, sales and transactions, plus a review system where users rate and comment on artworks.

• Movie Database : Movie Information Details

[Code](#)

Tools: Java, JavaFX

- Developed as a term project using Java and JavaFX, this project serves as a movie database that provides detailed information about movies. It allows users to search, view, and manage movie details through a user-friendly interface built with JavaFX.

• Side-Channel Attack : Website Fingerprinting

[Code](#)

Tools: Python, Flask, PyTorch, Selenium

- Implemented a cache-based side-channel attack that fingerprints websites by analyzing CPU cache timing patterns collected via a Selenium-automated browser pipeline, then trained two 1D CNN models in PyTorch to classify the target site without intercepting any network traffic.

• GradPilot

[Code](#)

Tools: Next.js, Spring Boot, PostgreSQL

[Demo](#)

- AI-powered platform that simplifies the graduate school application journey for Bangladeshi students: personalized university recommendations, verified mentors, AI-driven Statement of Purpose reviews, scholarship discovery and a community forum. Built as Spring Boot and React microservices, with Google Gemini powering the AI features.

• Collaborator

Tools: Next.js, TypeScript, Supabase, 100ms

[Code](#)

[Website](#)

- Real-time video meeting platform with teams, secure video calls, chat, file sharing, meeting recording, AI-powered transcription and automatic summaries. Built with Next.js and TypeScript, using 100ms for low-latency video and Supabase for authentication, real-time messaging, storage and team management.

RESEARCH & PUBLICATIONS

• Age-Aware Neurological Gait Pattern Discovery from Biomechanical Signals

Python, XGBoost, NMF, Signal Processing, Statistical ML

- Developed a 6-phase ML pipeline on a 188-subject multimodal gait dataset to separate age, walking speed, and neurological deficit in post-stroke gait. Built a polynomial aging manifold with bootstrap prediction intervals and three-pathway impairment z-scores to produce age-normalised severity profiles. Designed manifold-residual augmentation, a clinically interpretable synthetic gait generator validated by a leave-one-age-band-out falsification test, and exposed a leakage artefact via leakage-free nested cross-validation. Key findings: EMG-derived Synergy Complexity Index tracks clinical walking ability; ankle push-off is largely speed-mediated while knee-power absorption is a speed-independent neurological marker.

• Multi-Model Machine Learning Analysis of Ecological Load Ratio Using Classification, Regression, and Clustering Approaches in EU Countries

[Paper](#)

8th IEOM Bangladesh International Conference, Dhaka, Dec 2025 | Python, Scikit-learn, XGBoost, LightGBM, Optuna, PSO

- Co-authored a study of 27 EU countries (837 observations) predicting the Ecological Load Ratio (ELR) with seven tree-based classifiers and regressors and six clustering methods. Used PSO and genetic-algorithm feature selection (about 120 indicators down to 7), Optuna tuning and stratified K-fold cross-validation, reaching 0.987 accuracy (Gradient Boosting) and R^2 above 0.98 (AdaBoost, LightGBM). Population density and forest rents emerged as the strongest drivers.

TECHNICAL SKILLS

- **Programming Languages:** Python, Go, C++, TypeScript, JavaScript, SQL, C, Java, HTML, CSS
- **Backend & Systems:** Go, REST APIs, WebSocket, MQTT (Mosquitto), Binary Wire Protocols, mTLS / PKI, Docker, Linux, Tailscale, Private Fiber Network & IP Addressing
- **Embedded & Edge:** STM32 Nucleo and Jetson Orin Nano (host-side integration), Robot Operating System - ROS, Arduino UNO, ATmega32
- **Data Science & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Transformers, LangGraph, LoRA Fine-tuning, Numpy, Pandas
- **Web Development:** React, Next.js, SvelteKit, Node.js, Express.js, Spring Boot, Laravel
- **Cloud, DevOps & Tools:** AWS, Google Cloud (Vertex AI), Azure, Docker Compose, GitHub Actions, Vercel, Cloudflare R2, Git, Apache JMeter, Microsoft Excel & PowerPoint
- **Database Systems:** PostgreSQL, TimescaleDB, Oracle, Supabase, Appwrite

CERTIFICATIONS

Coursera: [Neural Networks and Deep Learning](#) · [Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization](#) · [Structuring Machine Learning Projects](#)

Kaggle: [Intro to Machine Learning](#) · [Intermediate Machine Learning](#) · [Pandas](#)

ADDITIONAL INFORMATION

Language: English (Fluent) **Interests:** Data Science, Automation, Artificial Intelligence and Problem Solving